

DutyDraft — Draft-Order Course of Action (COA) Comparison

Decision brief for the Commanding Officer · The Commandant's Own

Prepared by SNCOIC (GySgt Aird) · 2026-06-13 · Internal decision document

Purpose & the decision

DutyDraft schedules monthly duty by **fixed rules — not AI and not chance**. One policy decision remains: **how the monthly draft order is generated**. Weekend burden is already distributed **equally** across every Marine and is unaffected by this choice. This brief compares the four candidate methods using a **12-month simulation on the real 29-Marine roster** (July 2026 – June 2027).

Questions for command as you read:

- Can leadership explain *every* Marine's pick, every month?
- Does seniority still mean something?
- Is recent burden (weekends, double duty) actually compensated?
- Can a Marine end up at the bottom of the order *by chance* — and be told only “the draw”?

How this was tested (assumptions)

- The real roster and 12 consecutive real months were used; the same engine that runs the live draft generated every order.
- Weekend days each month were assigned **equally** (least recent burden first) — the system the unit already uses.
- **Double duty rotates up the seniority line**: when a month has more duty days than Marines, the most-junior Marines on rotation stand it (and pick first that month as compensation). No Marine is hit twice before all are hit once.
- The two random methods (COA 2 & 3) were each simulated 400× to report expected outcomes, not a single lucky draw.

Finding 1 — Weekend burden is equal, under every COA

Over 12 months every Marine stood between **2 and 4 weekends** (about 3.6 each) — a spread of just **2**. This is **identical under all four COAs**, because weekend assignment never touches the draft order. *The COA decision does not affect weekend fairness — that is already solved.*

Finding 2 — Explainability: how much of the order is chance

The key measure is the “**luck swing**”: on average, how many positions a Marine's monthly pick lands away from the order the rules alone would have given (rank + earned burden). **Zero means every pick is explainable.**

Method	Luck swing	What it means in practice
COA 1 — Weighted Seniority	0 spots	Every pick traceable to rank + earned burden. No chance.
COA 3 — Hybrid	2.9 spots	Bounded chance: only the top five slots are a draw.
COA 2 — Weighted Lottery	7 spots	Roughly 7 positions of pure chance layered on top of the rules.

Per Marine, per month, averaged over the 12-month simulation. COA 1 is the rules-based baseline, so it is 0 by definition.

Finding 3 — What it feels like to the Marine (trust & morale)

Consider **GySgt Aird (#3 most senior)** — same Marine, same earned burden, across the 12 months. Only the method changes:

Method	Monthly draft pick (Jul → Jun)
COA 1 — Weighted Seniority	8, 9, 3, 8, 6, 10, 6, 3, 8, 4, 7, 7
COA 2 — Weighted Lottery	21, 20, 10, 13, 3, 5, 24, 19, 19, 22, 14, 26

Under the rules-based order this senior Marine drafts in the single digits every month — exactly what his seniority earns. Under the lottery, **the same Gunnery Sergeant is dropped as deep as #26**, repeatedly, by chance. There is no reason to give him beyond “the draw.”

A rules-based order gives **every** Marine a reliable path to good picks: senior Marines through rank, junior Marines through their double-duty compensation month. A lottery removes that guarantee — a Marine can sit near the bottom month after month with no recourse and no explanation. **A duty tool the Marines do not trust will not be used.**

Double-duty rotation (climbs the seniority line, fairly)

Proof the compensation rotates and no one is hit twice before all are hit once:

Month	Stands double duty (→ picks first that month)
Jul 2026	LCpl McBride (#29), LCpl Sakamoto (#28)
Aug 2026	LCpl Walter (#27), LCpl Collins (#26)
Sep 2026	LCpl Arriaga (#25)
Oct 2026	Cpl Osterhout (#24), Cpl Cendan (#23)
Nov 2026	Cpl Ezeta (#22)
Dec 2026	Cpl Mashler (#21), Cpl Ramos (#20)
Jan 2027	Sgt Capone (#16), Sgt Gaskin (#15)
Feb 2027	— (short month)
Mar 2027	SSgt Johnson (#14), SSgt Bumgarner (#13)
Apr 2027	SSgt Mueller (#12)
May 2027	SSgt Miller (#11), SSgt Donaldson (#10)
Jun 2027	SSgt Jensen (#9)

One month, side by side — the real Jul 2026 draft

The actual order each method would produce next month. (×2 = double-duty Marine, two picks.)

Pick	COA 1 — Weighted Seniority	COA 2 — Weighted Lottery	COA 3 — Hybrid
1	LCpl Sakamoto (×2·t1)	LCpl Sakamoto (×2·t1)	LCpl Sakamoto (×2·t1)
2	LCpl Sakamoto (×2·t2)	LCpl Sakamoto (×2·t2)	LCpl Sakamoto (×2·t2)
3	LCpl McBride (×2·t1)	LCpl McBride (×2·t1)	LCpl McBride (×2·t1)
4	LCpl McBride (×2·t2)	LCpl McBride (×2·t2)	LCpl McBride (×2·t2)
5	GySgt Walker	SSgt Donaldson	SSgt Donaldson
6	GySgt McCreary	GySgt Walker	GySgt Walker
7	GySgt Steel	GySgt Steel	GySgt Steel
8	GySgt Aird	SSgt Weiland	SSgt Weiland
9	GySgt Pena	Sgt Hallam	Sgt Hallam
10	GySgt Menz	SSgt Mueller	GySgt McCreary
11	SSgt Donaldson	SSgt Miller	GySgt Aird
12	SSgt Miller	GySgt Menz	GySgt Pena
13	SSgt Jensen	SSgt Jensen	GySgt Menz
14	SSgt Weiland	Sgt Campa	SSgt Jensen
15	SSgt Kotasenski	Sgt Gaskin	SSgt Miller
16	SSgt Johnson	SSgt Johnson	SSgt Kotasenski
17	Sgt Campa	SSgt Kotasenski	Sgt Campa
18	Sgt Hallam	Cpl Ramos	SSgt Johnson
19	SSgt Mueller	GySgt Pena	SSgt Mueller
20	Sgt Rosie	GySgt McCreary	Sgt Rosie
21	Cpl Ramos	GySgt Aird	Cpl Ramos
22	SSgt Bumgarner	Sgt Rosie	SSgt Bumgarner
23	Sgt Gaskin	Cpl Mashler	Sgt Gaskin
24	Sgt Capone	LCpl Walter	Sgt Capone
25	Cpl Mashler	Cpl Ezeta	Cpl Mashler
26	LCpl Arriaga	SSgt Bumgarner	LCpl Arriaga
27	LCpl Collins	Cpl Osterhout	LCpl Collins
28	Cpl Ezeta	LCpl Arriaga	Cpl Ezeta
29	Cpl Cendan	Cpl Cendan	Cpl Cendan
30	Cpl Osterhout	LCpl Collins	Cpl Osterhout

Pick	COA 1 — Weighted Seniority	COA 2 — Weighted Lottery	COA 3 — Hybrid
31	LCpl Walter	Sgt Capone	LCpl Walter

12-month outcome per Marine

Average pick [best-worst] over the 12 months, by method, plus weekends stood. Sen = seniority rank.

Sen	Marine	COA 1	COA 2	COA 3	Wknds
1	GySgt Pena	4.8 [1-9]	14.3 [5-24]	9.1 [5-12]	4
2	GySgt McCreary	5.2 [2-7]	16 [8-23]	9.6 [7-12]	3
3	GySgt Aird	6.6 [3-10]	16.3 [3-26]	9.6 [3-13]	4
4	GySgt Steel	7.3 [4-10]	14.5 [4-26]	10 [4-14]	3
5	GySgt Walker	8.1 [5-11]	12.8 [2-27]	10.2 [2-16]	3
6	GySgt Menz	9.3 [5-12]	11.9 [5-21]	11 [5-15]	4
7	SSgt Weiland	10.3 [5-15]	15.3 [6-25]	11.3 [6-18]	4
8	SSgt Kotasenski	11.2 [7-15]	14.9 [3-25]	12.5 [3-17]	4
9	SSgt Jensen	11.3 [1-15]	14.4 [1-26]	11.2 [1-19]	4
10	SSgt Donaldson	11.8 [1-17]	14.1 [1-28]	11.2 [1-20]	3
11	SSgt Miller	12.3 [3-16]	17.3 [3-29]	13.8 [3-20]	3
12	SSgt Mueller	14.7 [1-19]	12.4 [1-29]	14.6 [1-21]	4
13	SSgt Bumgarner	14.2 [1-22]	12.8 [1-27]	12.3 [1-22]	4
14	SSgt Johnson	15.4 [3-21]	13.4 [3-26]	14.8 [3-19]	3
15	Sgt Gaskin	15.9 [1-23]	15.8 [1-27]	15.2 [1-23]	4
16	Sgt Capone	16.9 [3-24]	19.5 [1-31]	17.5 [1-24]	4
17	Sgt Hallam	19.3 [16-23]	16.5 [4-28]	17.2 [4-25]	4
18	Sgt Rosie	20.3 [17-23]	17.2 [5-30]	19.2 [5-25]	3
19	Sgt Campa	21.8 [17-25]	13.3 [3-28]	19.1 [3-24]	2
20	Cpl Ramos	20.3 [1-25]	13.5 [1-24]	17.1 [1-26]	3
21	Cpl Mashler	21.7 [3-27]	18.5 [3-29]	18.8 [3-27]	4
22	Cpl Ezeta	22.3 [1-28]	19.9 [1-31]	22.8 [1-28]	4
23	Cpl Cendan	22.8 [1-29]	19.6 [1-31]	21.5 [1-29]	4
24	Cpl Osterhout	24.2 [3-31]	20.3 [3-29]	23 [3-31]	4
25	LCpl Arriaga	23.9 [1-30]	21.6 [1-31]	24.2 [1-30]	3
26	LCpl Collins	25.3 [1-31]	18.5 [1-30]	21.9 [1-31]	3
27	LCpl Walter	25.5 [3-31]	21.8 [3-31]	23.9 [3-31]	4
28	LCpl Sakamoto	26.5 [1-31]	19.1 [1-31]	26.5 [1-31]	4
29	LCpl McBride	27.8 [3-31]	20.9 [3-30]	27.8 [3-31]	4

Scorecard & the decision

	COA 1	COA 2	COA 3
Every pick explainable to the Marine	Yes	No – “the draw”	Mostly
Seniority respected	Yes	Odds only	Yes
Recent burden compensated	Yes	Odds only	Yes
Outcome decided by chance	No	Yes (~7 spots)	Top 5 only
A Marine can land at the bottom repeatedly by luck	No	Yes	Only near the top

The choice is purely how much chance to introduce into the draft order — weekend burden is equal regardless. COA 1 maximizes explainability and trust; every pick can be defended to the Marine standing it. COA 2 maximizes opportunity for junior Marines, at the cost of explainability and seniority. COA 3 is a middle path: a rules-based order with a small, bounded lottery for the top five slots.

Generated from the DutyDraft draft-order engine and 12-month fairness simulation. All figures are live model output, not estimates. Seed-stable and reproducible.